COMP1551 Application Development
Education Centre Desktop Information System

Coursework, Term 2 2025/26 — Tasks 1 and 2

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Task 3 of this coursework is submitted separately as the single source file `EducationCentreSystem.cs`. The two diagrams in Task 2 describe the classes and the behaviour of that file. The same file is published at
https://github.com/ryanthemechanic/GRE-Coursework/blob/main/Year-2/Semester-2/COMP1551/EducationCentreSystem.cs

Task 1a — Description of the Desktop Information System

275 words.

The education centre keeps its records on paper. Teaching staff, administration staff and students each have a folder of forms, and the office copies a form by hand whenever somebody changes a telephone number or takes on a new subject. Finding one record means going through a drawer, and the same person often appears twice with different details.

For every person on the books the system holds a name, a telephone number, an email address and the role that person has at the centre. Each group then has details of its own. A member of the teaching staff has a salary and the two subjects they teach. A member of the administration staff has a salary, the hours worked each week, and whether the post is full time or part time. A student has the three subjects they study.

The office needs five things from the program: add a new person to the records; show every record held; show only the records of one group, so that the teaching staff can be listed without the students in the way; change the details of a person already recorded; and remove a person who has left the centre.

The program is for one member of the office staff at one machine. It should run on the centre's existing Windows 10 or Windows 11 desktops, which have 8 GB of memory and no graphics card beyond the one in the processor, and it should install without database software or a network connection. The office would rather type than use a mouse, so a numbered menu worked from the keyboard suits the way the staff already work.

Task 1b — Software Requirements Specification

378 words, excluding headings.

The structure and wording below follow Wiegers, K. and Beatty, J. (2013) *Software Requirements*. 3rd edn. Redmond, WA: Microsoft Press, chapters 10 and 11.

1. Introduction

Purpose. This document specifies the Education Centre Desktop Information System (ECDIS), version 1.0, which replaces the centre's paper records.

Project scope. ECDIS records the teaching staff, administration staff and students of one centre, and supports adding, viewing, filtering, editing and deleting those records. Timetabling, payroll and marks are out of scope.

2. Overall Description

Product. A self-contained desktop application. It connects to no other system and holds its records in memory while it is running.

Users. One records officer in the centre office, familiar with Windows but not with computing.

Operational environment. A Windows 10 or Windows 11 desktop with the .NET runtime, 4 GB of memory, no network and no database server.

3. System Features

Description. One record per person. Every record holds a name, telephone number, email address and role; teaching staff records add a salary and two subjects, administration records a salary, contract type and weekly hours, and student records three subjects.

Functional requirements.

- FR-1 The system shall present a menu of its operations and return to it when an operation ends.
- FR-2 The system shall add a record to one of the three groups, asking only for the fields that group holds.
- FR-3 The system shall refuse an empty or malformed name, telephone number, email address, salary or subject, and ask again.
- FR-4 The system shall list every record held with the fields belonging to it.
- FR-5 The system shall list the records of one group on request.
- FR-6 The system shall change any field of a chosen record, keeping the stored value where none is given.
- FR-7 The system shall delete a chosen record, after the user confirms.
- FR-8 The system shall store records in a structure whose size is not fixed in advance.

4. User Interface Requirements

A keyboard-driven text interface. Menu options are numbered, prompts name the field asked for, and while a record is edited its stored value is shown in brackets.

5. Platform Requirements

Windows 10 or Windows 11, 64-bit, with the .NET runtime. One executable, no installer, and nothing written to disk.

6. Quality Attributes

Performance. Any operation completes within one second for up to 1,000 records.

Security. The records are protected by the Windows account of the single office machine.

Safety. Deletion is confirmed and invalid data refused at entry, so a mistyped answer cannot corrupt the records.

Task 2a — UML Class Diagram

Figure 1 shows the classes of the system. `Person` is abstract and holds the data every record has; `Teacher`, `Admin` and `Student` inherit from it and add the fields of their own group, overriding `Role`, `PrintDetails` and `CaptureDetails`. `PersonRegister` owns the list that holds every record. `Validation` and `Prompt` are static classes: the first states the rule each field must satisfy, the second asks the questions at the keyboard and repeats them until the rule is met. Private members are marked −, protected members # and public members +.

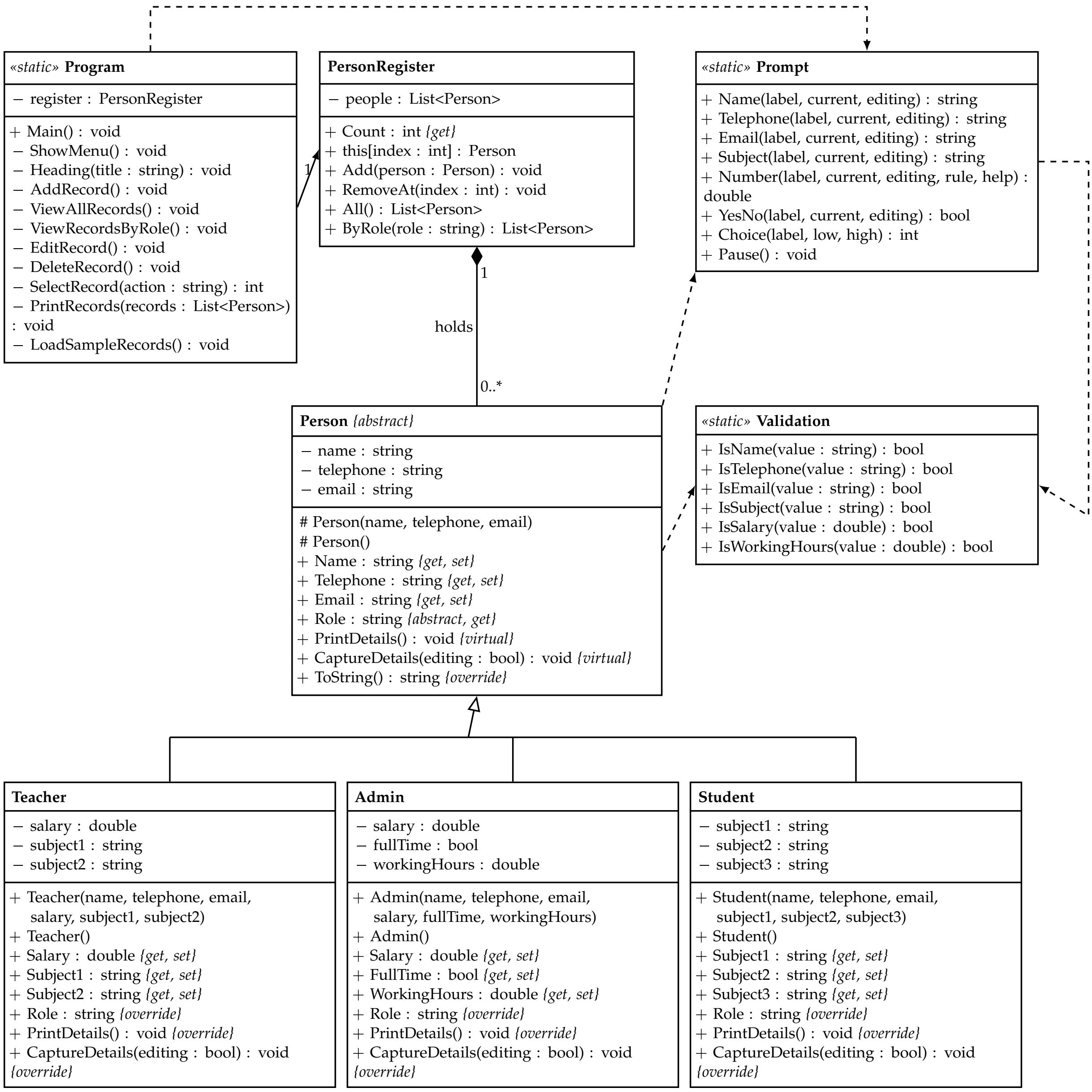

Figure 1: Class diagram of the Education Centre Desktop Information System.

Task 2b — UML Use Case Diagram

Figure 2 shows what the system does for the people who use it. The records officer runs all five operations from the menu; the centre manager uses the system only to read what is on it. Editing and deleting both begin by choosing a record from a numbered list, and adding and editing both check what has been typed before storing it, so those two steps are drawn as included use cases rather than repeated in each one.

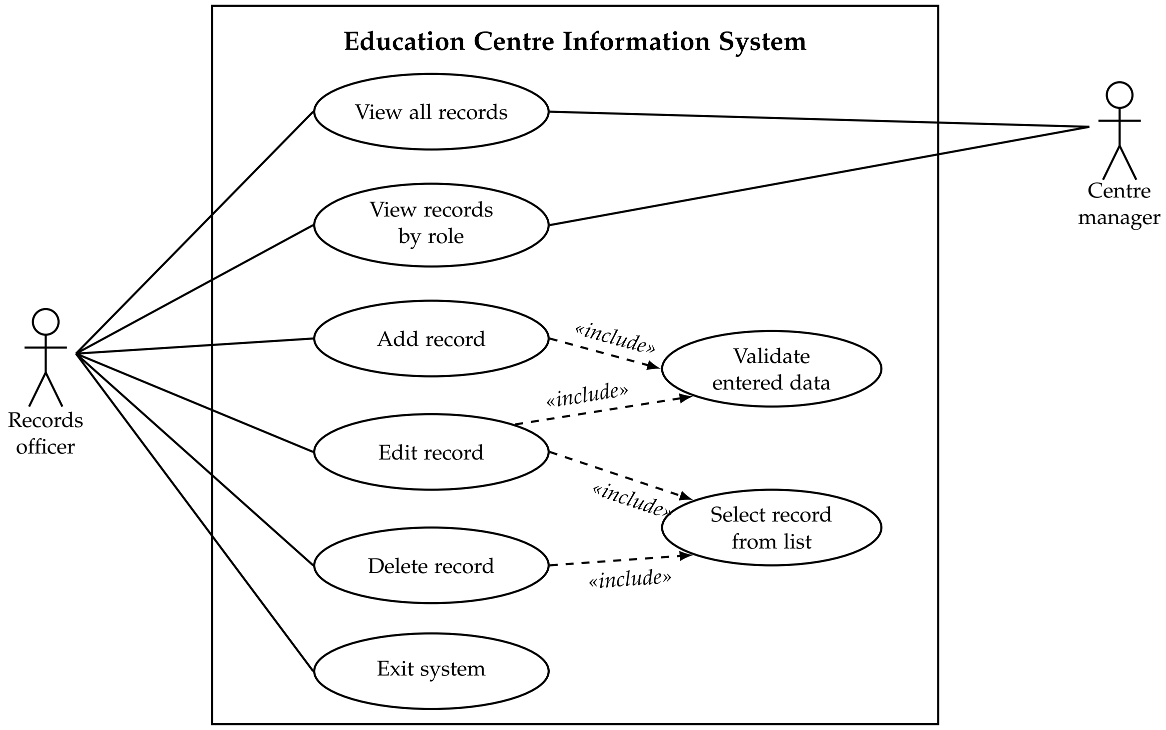

Figure 2: Use case diagram of the Education Centre Desktop Information System.